

Android ADB 壓力測試工具 (v4.0.4) 雙語操作手冊

Android ADB Stress Test Console (v4.0.4) Bilingual User Manual

文件版本 / **Version** : 1.0 (適用軟體版本 / For Software v4.0.4)

目標讀者 / **Target Audience** : QA 測試工程師、自動化測試人員 / QA Engineers, Automation Testers

壹、測試前準備與設備連線 / I. Pre-Test Preparation & Device Connection

在開始測試前，請務必確認設備環境，並將設備連線至本工具（強烈建議使用 WiFi 以避免線材鬆脫）。

Before starting any tests, please ensure the device environment is set up and connected to this tool (WiFi connection is highly recommended to avoid cable disconnection).

1. 開啟 USB 偵錯 / Enable USB Debugging

- **[中文]** 到手機的 設定 $\rightarrow$ 關於手機，連續點擊 版本號 7 次開啟開發者模式。接著到 設定 $\rightarrow$ 系統 $\rightarrow$ 開發人員選項，開啟 **USB 偵錯**。
- **[English]** Go to device Settings $\rightarrow$ About phone, tap Build number 7 times to enable Developer options. Then go to Settings $\rightarrow$ System $\rightarrow$ Developer options, and enable **USB debugging**.

2. 準備 OSD 浮水印檔案 (重要) / Prepare OSD Watermark File (Important)

- **[中文]** 請確保最新版的 TPM_OSD.apk (**v7.0**) 與本測試工具的執行檔放置於同一個電腦資料夾中。這能防止設備在長測時進入深度休眠。
- **[English]** Ensure the latest TPM_OSD.apk (**v7.0**) is placed in the **exact same folder** as this console executable. This prevents the device from entering deep sleep during long-term testing.

3. 設備連線指南 / Device Connection Guide

- **步驟 A：初始化無線連線 / Step A: Initialize Wireless Connection**
 - **[中文]** 先用實體 USB 傳輸線將手機接上電腦。點擊工具上的紫底按鈕「 **Init USB**」。看到 "Success" 提示後，請拔掉 **USB 線**。
 - **[English]** First, connect the phone to the PC using a physical USB cable. Click the purple " **Init USB**" button. After seeing the "Success" prompt, **unplug the USB cable**.

- **步驟 B：尋找並連線 / Step B: Scan or Connect**
 - **自動掃描 (Auto Scan)：**
 - **[中文]** 確認手機與電腦連在同一個 WiFi 下，點擊橘底按鈕「 **Scan**」，工具會自動找出手機並連線。
 - **[English]** Ensure the phone and PC are on the same WiFi network, click the orange " **Scan**" button. The tool will automatically find and connect to the device.
 - **手動連線 (Manual Connect)：**
 - **[中文]** 如果掃描不到，請去手機的 WiFi 設定看 IP 位址（如 192.168.1.50），輸入在 Manual IP 欄位，點擊藍色按鈕「 **Connect**」。
 - **[English]** If the scan fails, check the phone's IP address in its WiFi settings (e.g., 192.168.1.50), enter it in the Manual IP field, and click the blue " **Connect**" button.
- **步驟 C：選取設備 / Step C: Select Device**
 - **[中文]** 連線成功後，在中間的清單中**點選 (反白)** 您要測試的手機（支援多選）。
 - **[English]** Once connected, **click to highlight** the target device(s) in the middle dashboard list (multi-selection supported).

貳、全域設定與通用參數 / II. Global Settings & General Parameters

-  **Target (Cycles/Mins)**
 - **[中文]** 測試的**總長度**。若是 Monkey 隨機亂點或 CPU 燒機，代表「跑幾分鐘」；若是其他測試（如重啟、開啟相機、下載），代表「跑幾圈/次」。
 - **[English]** The **total duration** of the test. For Monkey random taps or CPU burn-in, this represents "Minutes"; for other tests (e.g., reboot, camera, download), it represents "Cycles/Times".
-  **Show OSD Watermark**
 - **[中文]** 勾選後，測試時會在手機螢幕印上大大的浮水印（設備序號、IP、目前測試），供 QA 即時監控。

- **[English]** If checked, a large watermark (Device Serial, IP, Current Test) will be displayed on the phone screen for real-time QA monitoring.

參、測試項目與專屬參數 / III. Test Categories & Parameter Details

1. 🚀 企業級 APM 核心測試 / APM Core Test Suites

(模擬企業商業環境中最常見的6種嚴苛操作 / Simulates the 6 most rigorous commercial operations)

- **[APM] System Restart & Shutdown Stress:**
 - **[中文]** 驗證硬碟開機分區壽命與電源管理。
 - **[English]** Verifies eMMC/UFS boot partition lifespan and power management.
- **[APM] Connectivity (WiFi/BT/Airplane) Toggle:**
 - **[中文]** 循環切換飛航模式、WiFi、藍牙，驗證射頻晶片與網路切換穩定度。
 - **[English]** Cyclically toggles Airplane mode, WiFi, and Bluetooth to verify RF chip and network switching stability.
- **[APM] Data I/O (Browser Download):**
 - **[中文]** 透過瀏覽器下載檔案，模擬真實大檔寫入。
 - **[English]** Downloads files via the browser to simulate real-world large file writing.
- **[APM] Burn-in (Video Streaming):**
 - **[中文]** 播放串流影片不間斷，驗證晶片散熱與高頻寬穩定性。
 - **[English]** Continuous video streaming to verify chip thermal dissipation and high-bandwidth stability.
- **[APM] Power & Display (Wake-up & Brightness):**
 - **[中文]** 喚醒螢幕，將亮度從最亮 (255) 降到最暗 (10)，驗證背光模組壽命。
 - **[English]** Wakes up the screen, adjusts brightness from max (255) to min (10) to verify backlight module lifespan.
- **[APM] Camera & Media Stress:**

- **[中文]** 啟動相機並觸發快門拍照，驗證影像處理器 (ISP) 是否有記憶體洩漏。
- **[English]** Launches the camera and triggers the shutter to capture photos, verifying ISP memory leaks.

2. 📷 相機與多媒體 / Camera & Media

- **Camera - Front/Rear:**
 - **[中文]** 切換前鏡頭拍 2 張照片，接著切換後鏡頭錄影 5 秒。
 - **[English]** Switches to the front camera for 2 photos, then to the rear camera for a 5-second video recording.
- **Camera - Continuous Shooting (100 shots):**
 - **[中文]** 模擬長按快門，連續極速連拍 100 張。
 - **[English]** Simulates holding the shutter for a high-speed continuous burst of 100 shots.
- **Camera - Switch Storage Space:**
 - **[中文]** 強制相機將檔案存在內建空間，下一圈再改存到記憶卡 (SD Card)。
 - **[English]** Forces the camera to save files to Internal Storage, then switches to External SD Card in the next cycle.
- **Audio - Playback & Controls Stress:**
 - **[中文]** 模擬真實使用者「播放、切換上下首、音量極值切換、暫停、快轉」。
 - **[English]** Simulates real user actions: "Play, Next/Prev track, Max/Min volume toggle, Pause, and Fast-Forward".
- **Audio - Background Play & Screen Lock:**
 - **[中文]** 音樂背景播放時鎖定螢幕 (待機 2 分鐘)，測試系統是否會錯誤關閉背景音樂。
 - **[English]** Plays music in the background, locks the screen (2 mins standby), and tests if the system improperly kills the background audio.

3. 💾 系統與儲存壓力 / System & Storage I/O

- **Storage I/O Stress (1GB dd):**

- **[中文]** Target Paths: 填寫要寫入大檔案的路徑。勾選 Test Concurrent 則會「同時」寫入多個路徑，極度耗效能。
- **[English]** Target Paths: Enter paths for large file writing. Check Test Concurrent to write to multiple paths "simultaneously" for extreme stress.

- **Storage Fake OOM Fill (%):**

- **[中文]** Target Fill Percentage: 要把手機硬碟塞到多滿 (預設 95%)。Hold Duration: 塞滿後維持該狀態幾分鐘。
- **[English]** Target Fill Percentage: How full to stuff the disk (Default 95%). Hold Duration: How many minutes to hold this full state.

- **Local File Copy Stress:**

- **[中文]** 設定來源 (Source) 與目的地 (Destination)，高壓複製檔案並自動刪除。
- **[English]** Set Source and Destination. High-stress file copying with automatic deletion.

- **Download Multiple / Large Files via WiFi:**

- **[中文]** Concurrent Files: 同時下載幾個檔案，嚴格考驗 WiFi 併發頻寬。
- **[English]** Concurrent Files: How many files to download simultaneously, rigorously testing WiFi concurrent bandwidth.

- **CPU Thermal Throttling (Mins):**

- **[中文]** 將 CPU 逼上 100% 滿載，驗證過熱降頻機制。
- **[English]** Pushes CPU to 100% load to verify thermal throttling mechanisms.

4. **應用與框架 / Apps & Framework**

- **Batch APK Installation Stress:**

- **[中文]** 指定本機電腦的資料夾，工具會把裡面的 .apk 依序自動安裝進手機。
- **[English]** Specify a local PC folder; the tool will automatically install all .apk files within it sequentially.

- **Multi-App Background & One-Click Clean:**
 - **[中文]** 瞬間開啟 10 多個 App 丟到背景，發送「一鍵清理」指令，並測試系統有無當機凍結。
 - **[English]** Instantly launches 10+ apps into the background, sends a "One-Click Clean" command, and checks for system freezes.
- **MDM Framework Stress (Work Profile) ⚠ 高危險操作 / High Risk:**
 - **[中文]** 測試設備的工作空間切換。鐵則：手機必須「恢復原廠設定」，且「絕不可登入 Google 帳號」，否則絕對報錯失敗！勾選 Set Device Owner 讓工具自動授權。
 - **[English]** Tests Work Profile switching. **Golden Rule: The phone MUST be "Factory Reset", and "DO NOT log into Google accounts", otherwise it will absolutely fail!** Check Set Device Owner for auto-authorization.
- **Monkey (Specific App / System-wide):**
 - **[中文]** Tap Interval: 每次亂點的延遲毫秒 (預設 300ms)。Skip System Apps: 勾選後會跳出確認框，把設定、電話等系統 App 加進黑名單，保證 Monkey 絕不誤觸。
 - **[English]** Tap Interval: Delay between random taps (Default 300ms). Skip System Apps: If checked, a dialog prompts you to blacklist system apps (Settings, Phone, etc.) so Monkey never accidentally touches them.

肆、動態面板參數補充 / IV. Dynamic Sub-Frames

-  **Reboot & Shutdown Settings (重啟設定) / Reboot Settings**
 - **[中文]** Wait after boot: 開機後等待穩定秒數。Include Shutdown Phase: 若手機支援「通電自動開機」才可勾選，會加入關機斷電階段。
 - **[English]** Wait after boot: Seconds to wait for OS stabilization after boot. Include Shutdown Phase: Check only if the device supports "auto-boot on power", adding a full power-off phase.
-  **Video Streaming Settings (影片燒機) / Video Settings**
 - **[中文]** Hrs/Cyc: 每一圈連播幾小時。Pause(M): 播完讓設備黑屏冷卻幾分鐘。

- **[English]** Hrs/Cyc: Hours to play continuously per cycle. Pause(M): Minutes to cool down with screen off after playing.
- 🌙 **Screen ON/OFF Settings (休眠喚醒) / Screen Settings**
 - **[中文]** Sleep Duration: 關閉螢幕待機秒數。Wake Duration: 喚醒螢幕亮屏秒數。
 - **[English]** Sleep Duration: Seconds to stay in sleep/standby mode. Wake Duration: Seconds to keep the screen awake.

伍、測試執行與日誌收集 / V. Execution & Log Collection

1. ▶ **START Selected (開始測試 / Start Test)**

- **[中文]** 設定完成後點擊開始。工具右側的 Log 視窗會即時印出底層 ADB 指令與設備回傳結果。
- **[English]** Click to start after configuring. The Log window on the right will output underlying ADB commands and device responses in real-time.

2. □ **STOP Selected (強制停止 / Stop Test)**

- **[中文]** 隨時中斷測試。工具會自動清理殘留的 Monkey 任務或下載到一半的垃圾檔案。
- **[English]** Interrupts the test at any time. The tool will automatically clean up residual Monkey tasks or partially downloaded junk files.

3. 📁 **Open Logs (收集報告 / Collect Logs)**

- **[中文]** 測試結束（無論 PASS 或 FAIL），工具會自動從手機抓取一份最完整的 **Bugreport (.zip)** 與 **操作日誌 (.txt)**。點擊按鈕會打開電腦資料夾，只需將檔案交給 RD 即可！
- **[English]** Upon test completion (PASS or FAIL), the tool **automatically** extracts a complete **Bugreport (.zip)** and an **Operations Log (.txt)** from the phone. Click this button to open the PC folder and simply hand these files to the RD team!